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Palaeomagnetic Measurements on the Great Dyke, Southern Rhodesia

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Abstract

A series of serpentine, pyroxenite and norite samples from the Great Dyke were collected near Chingford, Lalapanzi and Selukwe. All the measurements were initially scattered, but upon demagnetization in a.c. fields a consistent grouping resulted in the case of the norite samples. The Curie point of the ferromagnetic phase in the norites was found using an automatically recording Chevallier balance.

The Great Dyke is intruded into the Pre-Cambrian Basement complex and is itself regarded as Pre-Cambrian in age. From the mean site direction of magnetization an ancient pole position was found to be 10·7 N 68·9 E.

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§ 1. Introduction

The Great Dyke is a feature of considerable geological interest, never exceeding seven miles in width, it strikes in a N.N.E.–S.S.W. direction with very little disturbance over the greater part of its length of over 300 miles. It is intruded largely into granites of the Pre-Cambrian Basement complex, and highly metamorphosed xenoliths of hornfels and banded ironstone belonging to the Basement complex have been found in it. The Dyke is thought to be post-lomagundi in age and from the general similarity of pseudo-stratification, the alternate banding of chromite, pyroxenite and serpentine as well as from the general strike, a genetic relationship with the Bushveld complex has been inferred. The age of the latter is known to be of the order of 2000 million years from radioactive age determinations reported by Schreiner (1958).

Petrologically the rock types represented in the Great Dyke range from ultra-mafic to mafic types and include serpentine dunite pyroxenite and gabbroic rocks such as anorthositic gabbro, norite, gabbro and quartz gabbro. Transitional rock types occur and in this category harzburgite, olivine pyroxenite and picrite are found.

The serpentine is a near surface phenomenon, for it is not recorded below a depth of 1000 ft except in the neighbourhood of faults. It is believed to be predominantly of autometasomatic origin due to the effects of late magmatic fluids from the parent melt on consolidated, but still hot, olivine (dunite) rocks. It may still occur at the present day, due to the circulation of meteoric water in the vicinity of fault planes. Magnetite which has been recorded as abundant, although as will appear later the

amount would appear to vary considerably, is probably secondary. The same is true of the magnetite found in the pyroxenite bands. The pyroxenite is a largely unaltered monomineralic rock with interstitial olivine and plagioclose sometimes occurring.

According to Worst (1958) the parent magma of the Great Dyke was poor in iron and no marked concentration of iron in the form of magnetite occurred. The iron-rich end-phase magma is represented by comparatively iron-rich pyroxene in the gabbroic rocks, where magnetite also is found. These petrological conclusions are fully borne out by the magnetic measurements.

During the summer of 1958 a series of samples was collected representing some of the petrological types present in the Great Dyke. The area sampled is covered by Geological Survey of Southern Rhodesia Bulletins 39 (Tyndale-Biscoe 1949) and 44 (Wiles 1957) in which the dyke is described. The samples come from the Hartley and Selukwe Complexes of Worst (1958). Localities were visited with R. Tyndale-Biscoe and A. E. Phaup, to whom grateful acknowledgment must be made for their help. The interest of Professor S. K. Runcorn and Dr. K. A. Davies at all stages was appreciated, and thanks are due to the manager and management of the Cambrai Chrome mine for facilitating the collection of samples underground.

Concurrent with this investigation, research has been carried out in Salisbury (McElhinny and Gough 1963). Although using different techniques the results are substantially the same.

§ 2. Field Sampling

Over 50 orientated samples of the Great Dyke were collected from 20 sites. A series of five sites in the norite were collected near Chingford. The most westerly site was on the banks of the Serui river, with the remaining four within about three miles to the east and close to the road. A group of three samples was also obtained from a hill-top site bordering the Churu road near Selukwe. The former lie in the Hartley complex of Worst, the latter in the Selukwe complex.

Associated with the Great Dyke and lying to the east and the west of it are a series of bordering dykes of which one, the Umvimeela dyke was sampled near the railway line in the vicinity of Guinea Fowl siding, and at a further site a dyke, presumed to be either the same or an associated dyke, exposed in a road cutting north-east of Guinea Fowl siding and near the aerodrome was collected.

Pyroxenite and serpentine samples were collected in the Lundi river near Pons Chrome mine, Selukwe, in the Little Umtebekwe river, from the workings of the Cambrai Chrome mine, Lalapanzi and from near the York Chrome mine east of Que Que.

At each site two or three orientated samples were collected from approximately the same horizon, i.e. with respect to the stratification of the dyke.

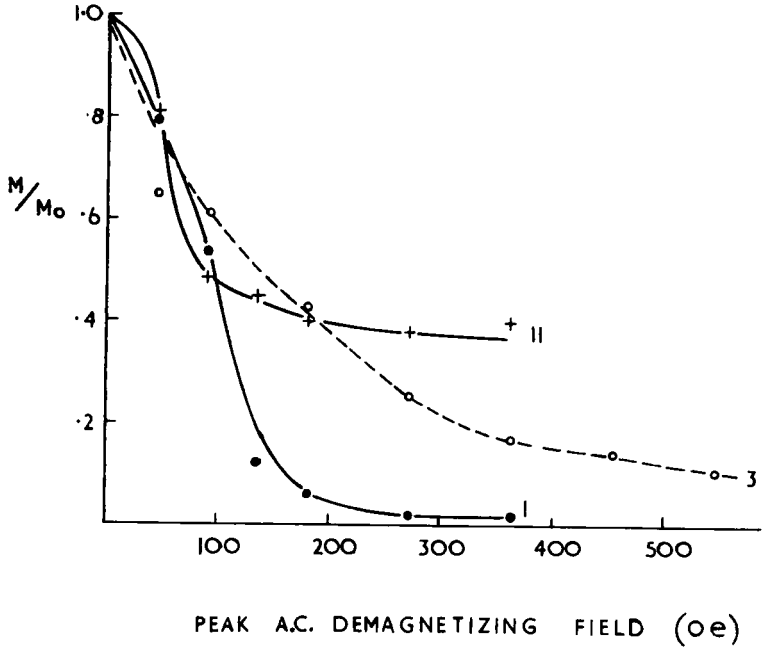
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§ 3. Measurement and Demagnetization

From each sample two 1 in. cores were prepared and from these up to four 1 in. cylinders per sample.

The direction and intensity of natural remanent magnetism of each core was measured under a short-period igneous magnetometer. When the directions are plotted site by site a pattern of near-random directions, characteristic of what is known of remagnetization by lightning, emerges. Great circle streaking due to a viscous component of magnetization was absent. In such cases it has been shown that secondary components may be removed by a.c. demagnetization (Graham 1961, Cox 1961) and consequently all samples were so treated.

Fig. 1



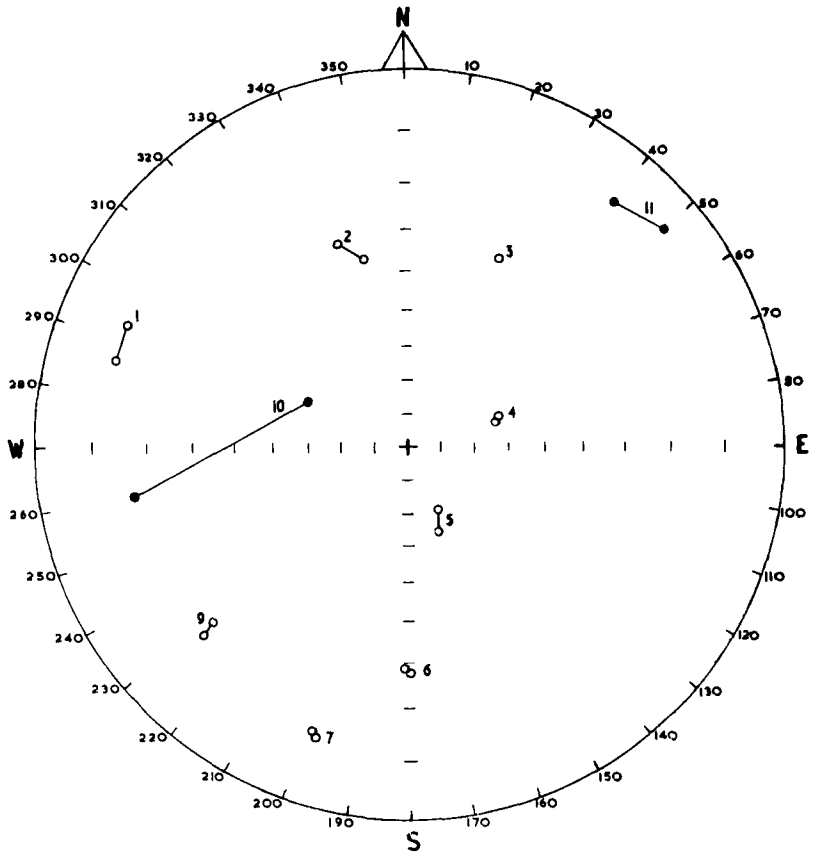
A.C. demagnetization curves of norite samples from the Hartley complex, Chingford.

Initially about four cores per site were progressively demagnetized in fields of increasing magnitude whilst being rotated about two axes in the zero field space (cancellation to less than 50 γ produced by a set of square coils. Typical demagnetization curves of norite samples are illustrated in fig. 1. Smooth field reduction was obtained by using a small electric motor driving a conical pulley to withdraw one electrode from a 1·5 m electrolyte tower (see Creer 1959).

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It is the common experience in a.c. demagnetization that the scatter of the directions of magnetizations is at first reduced, this being associated with greater or lesser changes in direction. As the field intensity increases, the scatter again increases although the mean direction may not change significantly until high fields are applied. The interpretation suggested by Irving *et al.* (1961) is that a viscous magnetization due to magnetic particles with a relaxation time short compared with the age of the rock is removed by fairly low fields (in many cases 150 oe is sufficient). When this is removed the residual magnetization is believed to be the primary one acquired at the time the rock was formed, and this direction changes very little although the scatter increases until at high fields, when the rock is nearly demagnetized, the site directions approach a random distribution due to the demagnetization procedure. The optimum

Fig. 2



Stereographic plot of directions of natural remanent magnetization in the norite samples from the Hartley complex, and in samples from one of the associated smaller dykes. Cores from the same sample are joined together (no more than two per sample plotted).

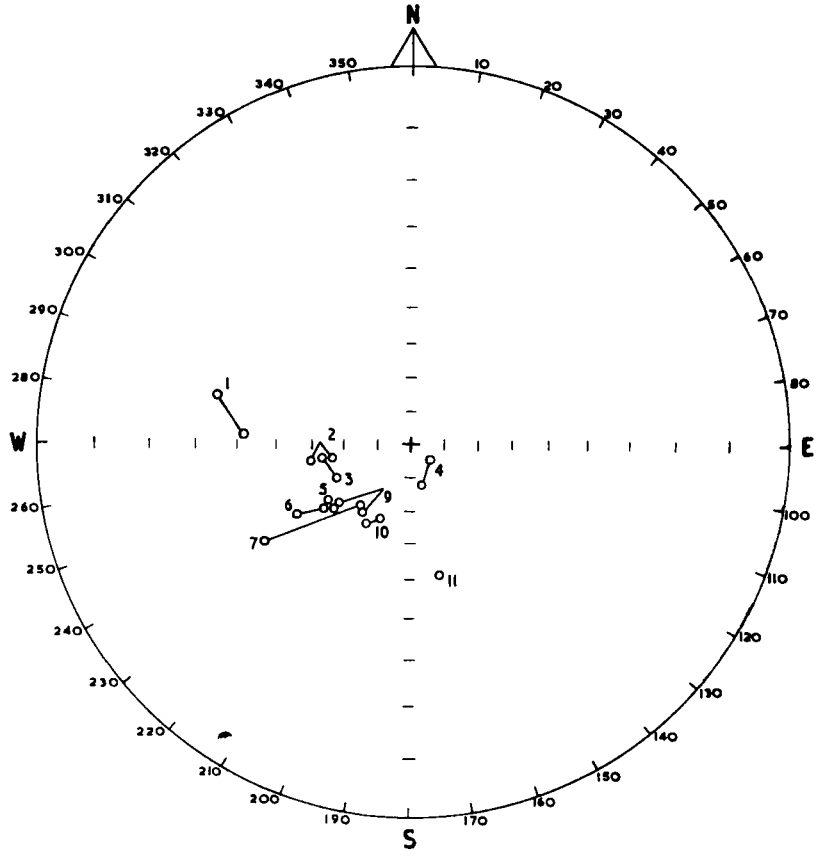
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demagnetizing field for any given site was considered to be that which gave a minimum scatter and which resulted in a consistency of directions of magnetization from different sites in the same formation or of rocks of the same age.

Adopting these criteria only the norite samples from the Hartley complex near Chingford and the samples from the bordering dyke responded to a.c. treatment, no consistent results being obtained from the pyroxenite and serpentine samples. The results from the norite samples can be illustrated by reference to figs. 2 and 3. In the present case the optimum field for two sites upon progressive demagnetization was found to be 300 oe, whereupon all the remaining cores were demagnetized at 250, 300 and 350 oe. The site minimum scatter could then be found by computing site means using all cores.

At two sites however one sample did not show any appreciable direction change whilst the other conformed to the general pattern. In both cases

Fig. 3



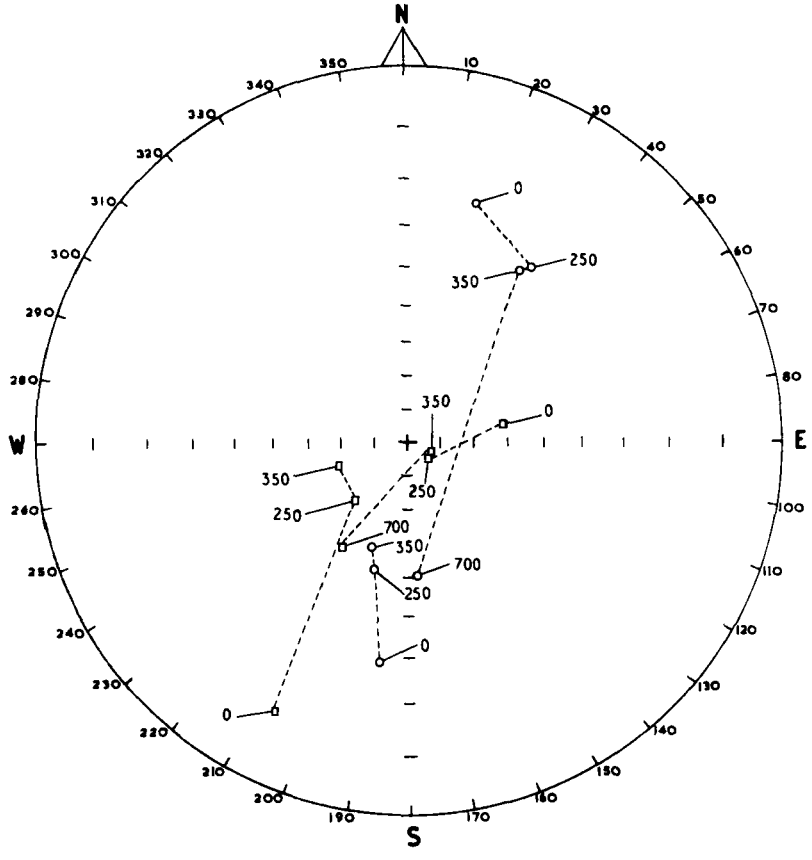
Stereographic plot of the directions of magnetization after demagnetization. Sample numbers are the same as in fig. 2.

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the intensity of magnetization of the sample showing small change in direction was anomalously high. Graham (1961) has shown both that the secondary magnetization due to lightning may be magnetically quite hard, and also that the effects of lightning are comparatively local. Consequently demagnetization of these specimens was continued until at a field of about 700 oe the directions changed to conform to the other samples; this is illustrated in fig. 4. The range of the residual intensity of magnetization at this point irrespective of the demagnetizing field was from 1 to 4×10^{-4} e.m.u./cm^3, whereas the initial natural remanent magnetization lay in the range of 2 to 78×10^{-3} e.m.u./cm^3. This result would suggest that care is required in applying standard demagnetization routines to rocks where lightning effects are suspected.

The site mean directions were given unit weight in the computation of an overall site mean from which the ancient pole position could be found.

Fig. 4

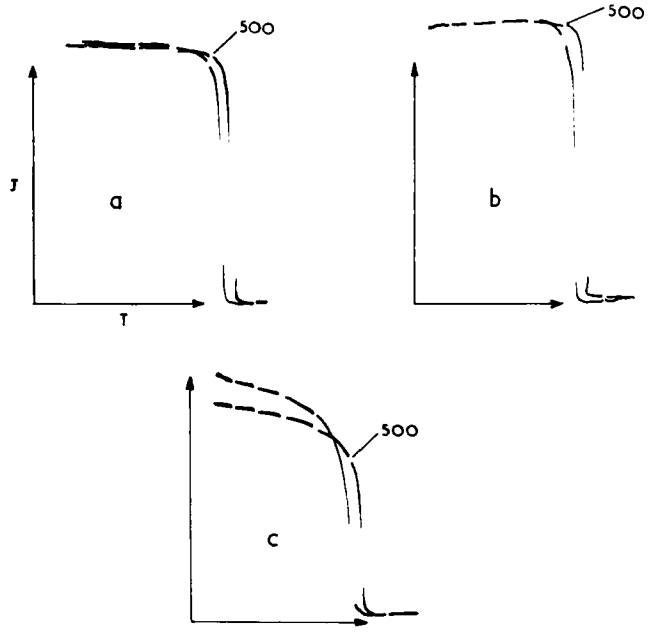


Stereographic plot of changes in directions of magnetization at sites 2 (denoted by open circles) and 3 where demagnetizing fields of up to 700 oe were required for one sample from each site. The peak a.c. demagnetizing fields are shown on the figure.

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A number of thermo-magnetic curves were made on a modified Chevallier balance, built by Parry (1957), in order to determine the Curie point of the magnetic phase. Specimens were mounted on a torsion arm lying between the poles of an electromagnet and movement of this arm reflects changes in the magnetic properties of the specimen as it is heated, the balance deflection falling to zero at the Curie point. The electromagnet gave inhomogeneous fields of up to 1650 oe although, in general, lower fields were applied. The value of the field to which the specimen was subjected was closely dependent upon position and consequently varied during the course of experiment with the movement of the torsion arm.

Fig. 5



Thermo-magnetic curves obtained from a modified, automatic, Chevallier balance. The drawings are copies of the photographic record. The curves record the change in intensity with temperature, every 100°C temperature interval being marked by a break in the trace. The upper curve represents the initial heating, the lower, the cooling curve. (*a*) magnetite sample (max. field 1000 oe); (*b*) norite sample from site 4 (max. field 450 oe); (*c*) norite sample from site 5 (max. field 1200 oe).

Copies of two photographic records made on norite samples from sites 4 and 5 (fig. 5 (*b*), (*c*)) are typical and a single magnetic phase with a composition close to that of pure magnetite is suggested by the Curie point in the range of 530–540°C. Curve 5 (*a*) represents a magnetite sample for comparison.

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§ 4. Discussion

Implicit in the computation of an ancient pole position from the mean site direction are the assumptions that in obtaining the mean site direction the effects of secular variation have been averaged out, and that the earth's magnetic field has always approximated to that of a geocentric axial dipole. Following Jaeger's (1957) arguments on the time required for the cooling of igneous bodies, the time represented by the six sites probably spans several thousands of years. It can be argued that the number of sites is inadequate to eliminate secular variation effects, and this may account in part at least for the difference between the mean and that reported by McElhinny and Gough (1963), although the difference is not significant. The results obtained are tabulated below.

Table 1. Site mean directions of magnetization

Site		Mean declination	Mean inclination	A.C. field applied	Samples per site
Hartley norite	1	222·8	−58·3	350 oe	1†
	2	181·2	−56·3	300 and 700 oe	2
	3	217·1	−75·5	250 and 700 oe	2
	4	283·1	−34·6	300 oe	1‡
	5	231·6	−57·8	300 oe	2
Umvimeela dyke		247·0	−55·7	250 oe	2
Overall mean site direction		236·3	−61·0	$\alpha=19{\cdot}9$, $K=12{\cdot}3$, $R=5{\cdot}593$	

† One sample excluded because of suspected field orientation error.

‡ One sample did not respond to a.c. treatment even in the highest field applied.

α = radius of circle of confidence, K = a measure dispersion, R = length of resultant vector mean from the six sites.

Bearing this in mind it is interesting to compare the results of this investigation with those of McElhinny and Gough (1963) and that of Gough and van Niekerk (1959) for the Bushveld gabbro.

Table 2. Pole positions of the Great Dyke and Bushveld complex

Great Dyke (McElhinny and Gough 1963)	21½ N	61½ E	$\alpha_{95}=9°$†
Great Dyke (this paper)	10·7 N	68·9 E	$d\psi=23{\cdot}3$ $d\chi=30{\cdot}5$
Bushveld gabbro (Gough and van Niekerk 1959)	23 N	36 E	$\alpha_{95}=12°$†

† α_{95} value given as poles were calculated as a mean of site poles.

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